

- **Storage Node (Personal Cloud)**
 - **Case:** Jonsbo N4 (Wood/Mesh finish) - supports up to 6x 3.5" drives and 2x 2.5"
 - **Motherboard:** ASRock N100M (Integrated Intel N100 CPU) - efficient for 24/7 operation.
 - **Memory:** 16GB DDR4 3200MHz - maximizes capacity for ZFS or specialized file caching.
 - **Boot Drive:** 128GB Sata SSD - keeps all SATA lanes dedicated to storage array.
 - **Primary Array:** 4 x 512GB Micron SATA SSDs - for a high-speed RAID 5/6 personal cloud.
 - **Bulk Storage:** 4 x 1TB 3.5" Hard Drives - raid5 for media and redundant lab backups.
 - **Power Supply:** Corsair SF750- required for the Jonsbo N4's compact layout.
- **Network Node (Firewall & Routing)**
 - **Chassis:** Lenovo ThinkCentre M920q Tiny.
 - **Processor:** Intel Core i5-8500T (6 Cores).
 - **Memory:** 16gb DDR4 2666MHz ram
 - **Storage:** 256GB NVMe SSD
- **Lab Node (Compute Cluster)**
 - **Chassis:** HP EliteDesk 800 G6
 - **Processor:** i7-10700
 - **Memory:** 40GB DDR4 2666MHz
 - **Storage:** 512GB NVMe SSD
- **Infrastructure**
 - **Switch:** NetGEAR GS308E
 - **3D Printer:** Elegoo Centauri Carbon 1 - used for printing custom 1 server rack, mount brackets, SSD spacers, etc
 - **Filament:** PETG-CF (Carbon Fiber) - provides the structural rigidity and heat resistance needed for internal server components. ASA also available.
 - **Cabling:** Cat6 shielded ethernet cables